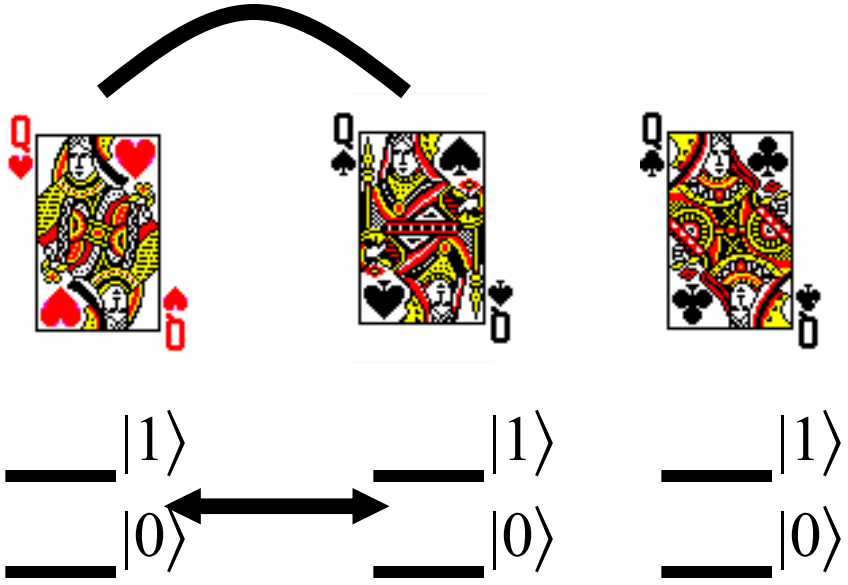
Quantum Three Card Monte



Dave Bacon

University of California Berkeley

Whaley Group

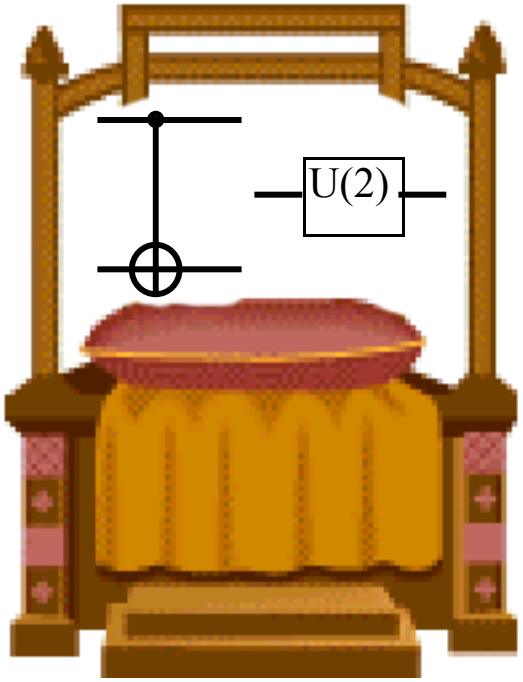
Cult of Universality

Universal Quantum Computer

≡ a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

≡ a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

An Engineer and a Physicist are stranded on a deserted island...

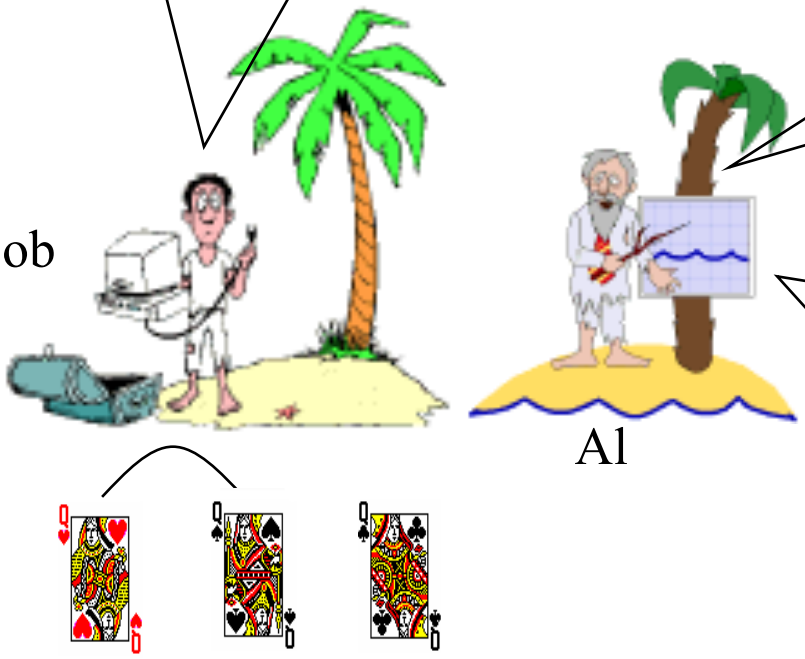
Look, Al, I've got this quantum system in my lab where I can turn on and off such-and-such an interaction and perform such-and-such procedures. **Can I quantum compute?**

Hmmm…..

Well, Bob, your such-and-such and interactions are not CNOT plus single qubit rotation on you qubits.

But…..?!

Bob



Al

Lets talk about...

DECOHERENCE-FREE SUBSPACES and DECOHERENCE-FREE SUBSYSTEMS (DFSs)

Collective Operations

$$s_x = \tfrac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad s_y = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad s_z = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \left[s_\alpha, s_\beta \right] = i\varepsilon_{\alpha\beta\gamma} s_\gamma$$

Total angular momentum operators on n qubits → Collective su(2) operators:

$$S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)} \text{ where } s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I \qquad \left[S_\alpha^n, S_\beta^n \right] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$

$$S_x^2 = \frac{1}{\sqrt{2}} \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_y^2 = \frac{1}{\sqrt{2}} \left(\begin{array}{ccc|c} 0 & -i & 0 & 0 \\ i & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_z^2 = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

triplet states $\begin{cases} |J=1, m_J=1\rangle = |00\rangle \\ |J=1, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \\ |J=1, m_J=-1\rangle = |11\rangle \end{cases}$ singlet state $\{ |J=0, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$

Intro quantum mechanics: addition of angular momentum for two spin-1/2 particles.

- Two spin-1/2 can be combined into a spin-1 triplet and a spin-0 singlet.

(Bob is a bit insulted by Al. He remembers this from QM 101.)

Degeneracy

Something special happens with three particles…

•spin-1 (triplet) plus spin-1/2 can produce either a spin-1/2 or a spin-3/2.

•spin-0 (singlet) plus spin-1/2 can produce spin 1/2.

$$S_x^3 = \left(\begin{array}{cccc|cc|cc} 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{\frac{3}{2}} & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \end{array}\right)$$

$$\left\{\begin{array}{l} \left|J=\tfrac{3}{2}, m_J=\tfrac{3}{2}\right\rangle = |000\rangle \\ \left|J=\tfrac{3}{2}, m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(|001\rangle+|010\rangle+|100\rangle\right) \\ \left|J=\tfrac{3}{2}, m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(|011\rangle+|101\rangle+|110\rangle\right) \\ \left|J=\tfrac{3}{2}, m_J=-\tfrac{3}{2}\right\rangle = |111\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle-|10\rangle\right)|0\rangle \\ \left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle-|10\rangle\right)|1\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}|00\rangle|1\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle+|10\rangle\right)\right\}|0\rangle \\ \left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}|11\rangle|0\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle+|10\rangle\right)\right\}|1\rangle \end{array}\right.$$

These spin-1/2 states come from combining a spin-0 two qubit state and a spin-1/2 single qubit state

These spin-1/2 states come from combining a spin-1 two qubit state and a spin-1/2 single qubit state

There is a **DEGENERACY** in the action of the 3 qubit collective su(2) operation!

Collective Decoherence



system-bath Hamiltonian

Bath mode with a wavelength much larger than the inner qubit spacing

$$H_I = \sum_{i=1}^{n} \sum_{\alpha=\{x,y,z\}} c_{\alpha,i}\, \underbrace{s_\alpha^{(i)}}_{\text{system operator}} \otimes \underbrace{b_{\alpha,i}}_{\text{bath operator}} \quad \overset{c_{\alpha,i}=c_\alpha}{\Rightarrow} \quad H_I = \sum_{\alpha=\{x,y,z\}} \sum_{i=1}^{n} c_\alpha s_\alpha^{(i)} \otimes b_\alpha = \sum_{\alpha=\{x,y,z\}} S_\alpha^n \otimes B_\alpha$$

$$S_\alpha^3 = \left(\begin{array}{c|c:c} s_\alpha^4 & 0 & 0 \\ \hline 0 & s_\alpha^2 & 0 \\ \hdashline 0 & 0 & s_\alpha^2 \end{array}\right) \quad \text{where } s_\alpha^k \equiv \begin{cases} \text{the k-dimensional irreducible} \\ \text{representation of su(2)} \end{cases}$$

S_x^3 over the degenerate portion $= \left(\begin{array}{cc:cc} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{array}\right) \rightarrow I_2 \otimes s_x^2$

$$S_\alpha^3 \rightarrow s_\alpha^4 \oplus \left(I_2 \otimes s_\alpha^2\right)$$

-action of S_α^3 on subspace which has the degeneracy acts as identity on a tensor component!

-prepare state with support only over the degenerate subspace which mimics the tensor product structure: system density matrix $= \rho_S = 0_4 \oplus \left(\omega_2 \otimes \pi_2\right)$.

-info in ω_2 will NOT decohere! info in π_2 will decohere! (think of $I \otimes \sigma_\alpha$ for two qubits)

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Decoherence-Free Subsystems (KLV)

General system-bath Hamiltonian $H_{SB} = \sum_\alpha A_\alpha \otimes B_\alpha$

A_α: system operators; B_α: linearly independent bath operators

A_α form an algebra A (add, multiply them) which is decomposable into subsystems:

$$\mathrm{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

M_{d_J}: linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator

n_J is the degeneracy of the Jth irrep

Quantum information encoded into a single degenerate component subsystem will not decohere! Decoherence-Free Subsystems (DFSs)

When $d_J = 1$ the information will be encoded over a subspace: Decoherence-Free Subspaces

"The universe is an enormous direct product of representations of symmetry groups" - Steve Weinberg



Bob

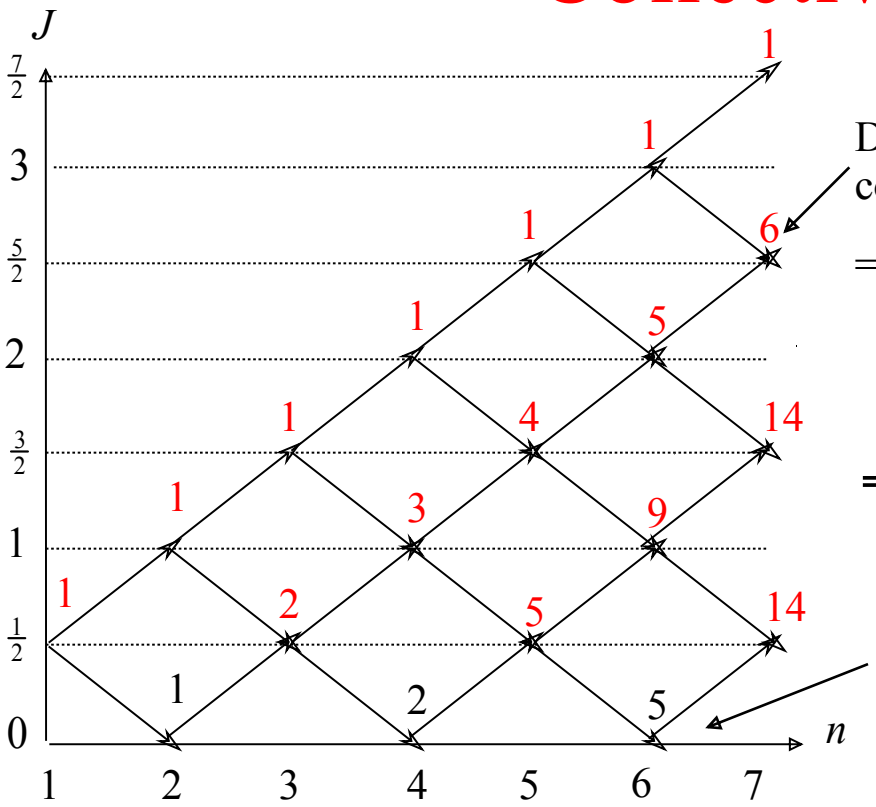
Great, Al! This is a neat possible way of avoiding decoherence! But what does this have to do with my question: will my system be able to quantum compute?

Hold your horses, Bob! These subsystems are only useful if we can compute on them…then I think you'll get my gist...

Al



Collective DFSs



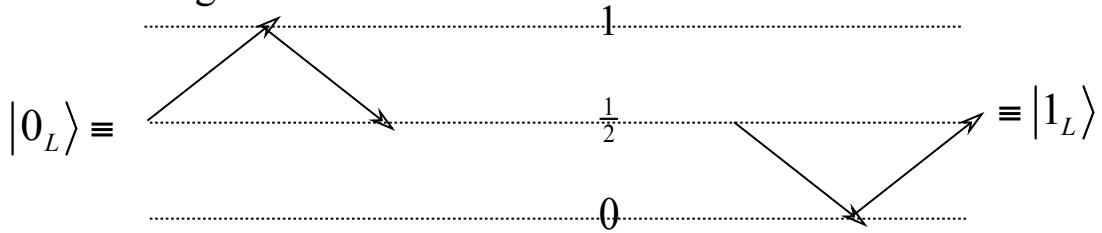
Degeneracy of the n qubit, total angular momentum J collective su(2) operators

= dimension of quantum information which can be encoded into the corresponding DFS.

$$= \frac{(2J+1)n!}{\left(\frac{n}{2}+J+1\right)!\left(\frac{n}{2}-J\right)!}$$

The J=0 irreps of su(2) are 1 dimensional Decoherence-Free Subspaces.

A nice set of basis states for a (n,J) DFS can be labeled by the path taken when constructing the (n,J) DFS via addition of angular momentum.



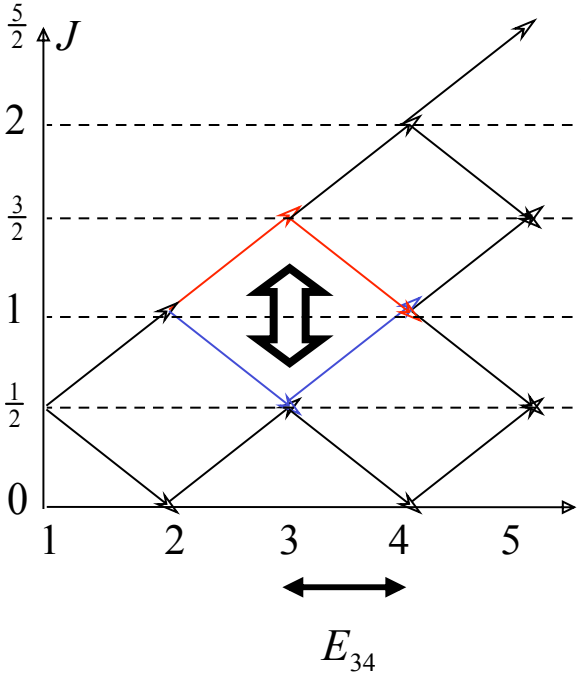
How do we perform quantum computation on the information of a (n,J) DFS?

Can we do this using only one or two-qubit interactions?

Exchange

The Exchange Hamiltonian between 2 qubits $\equiv E = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$. $E_{ij} \equiv$ E acting between qubits i and j.

Exchange commutes with the collective su(2) operators: $\left[E_{ij}, S_{\alpha}^{n}\right] = 0$ if $i, j \le n$



For a fixed number of qubits n (i,j ≤ n), E_{ij} does not mix (n,J) DFSs because E_{ij} commutes with the total angular momentum operator:

$\left[E_{ij}, \left(S^{n}\right)^{2}\right] = 0$ where $\left(S^{n}\right)^{2} = \left(S_{x}^{n}\right)^{2} + \left(S_{y}^{n}\right)^{2} + \left(S_{z}^{n}\right)^{2}$

But, E_{ij} does mix the different basis states!

Think: it changes the order in which the addition of angular momentum is performed.

Exchange is naturally fault tolerant on the DFSs!

The action of the exchange never exposes the info to the DFS decoherence.

The Exchange Hamiltonian between qubits 3 and 4, E_{34}, mixes the red and blue pathway basis states.

Exchange's Universality

Consider the ability to use the exchange Hamiltonian between qubits (nearest neighbors is enough)

$$U_i(t) = \exp\left[\frac{i\lambda E_{i,i+1} t}{\eta}\right] \quad \lambda = \text{coupling constant}, \quad i = 1 \ldots n-1$$

or, even just the ability to perform the gate $U_i(t)$ for $\lambda t/\eta$ an irrational multiple of π.

This set of gates is universal on each individual (n,J) DFS! (and not more!)

The exchange Hamiltonian is universal on the encoded states of a collective DFS

Julia Kempe, Dave Bacon, Daniel Lidar, K. B. Whaley, "Theory of Decoherence-Free Fault-Tolerant Universal Quantum Computation." quant-ph /0004064. (2000)

Simple example for the $n = 3$, $J = \frac{1}{2}$ DFS:

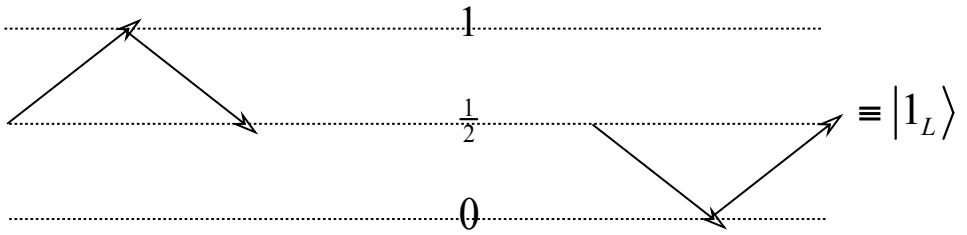
recall that the states are formed from combining a triplet with a spin-$\frac{1}{2}$ state or or by combining a singlet with a spin-$\frac{1}{2}$ state. Triplet is invarient under exchange and singlet flips sign under exchange. Thus

$E_{12} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ on the logical qubits.

more details tell us that

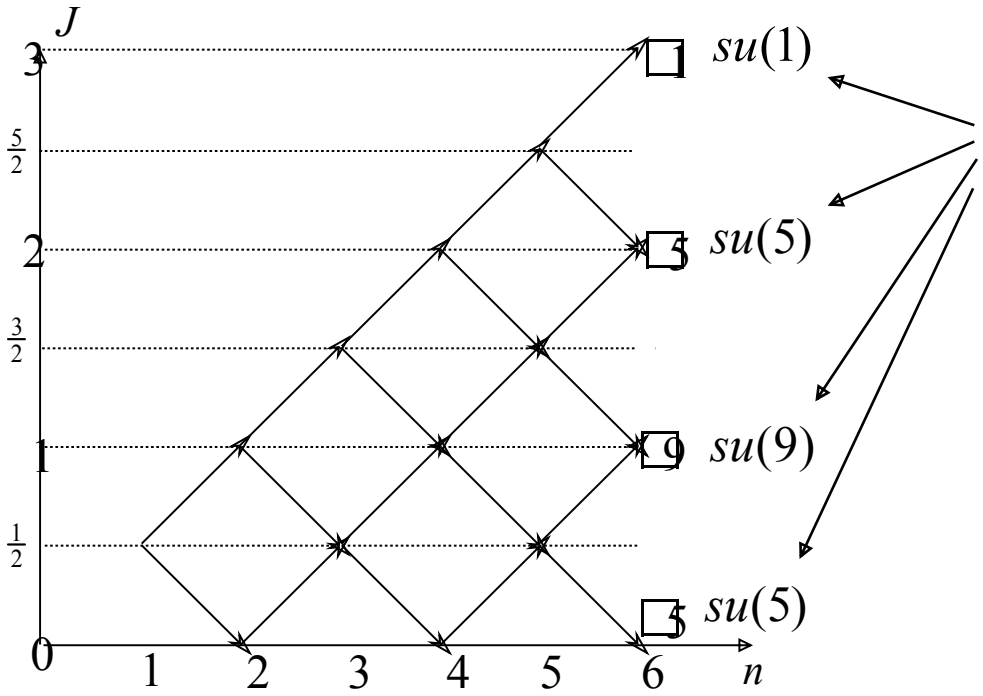
$E_{23} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ on the logical qubits.

$|0_L\rangle \equiv$



$\equiv |1_L\rangle$

Universality Amazing!



Can perform su(d)'s on each of the DFSs

encode information into the DFSs and the EXCHANGE interaction can be used to perform any su(d) on each of the su(d)'s for a given n independently!

su(d) means we have a d dimensional Hilbert space (encoded) and we can perform all unitary manipulations on this d dimensional Hilbert space!

$$\text{encoding efficiency} \equiv r = \frac{\log_2(\text{dimension of DFS})}{\text{n}}$$

As $n \to \infty$, $r(n) \to 1 - c\dfrac{\log_2 n}{n}$. The encoding efficiency is nearly unity for large n!

The Exchange Hamiltonian is ASYMPTOTICALLY UNIVERSAL

Issues

- Concatenation of collective DFSs.
- Leakage detection between collective DFSs.
- Fault-tolerant measurement on collective DFS states.
- Quantification of gate times required to perform actual operations.
- For fixed J, quantum computation and computation on collective DFSs are computationally equivalent for large n (qc⇒dfs trivial, dfs⇒qc a bit harder)

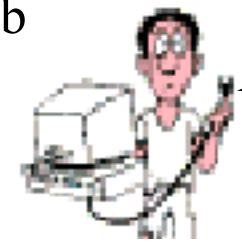
Dave Bacon, Julia Kempe, Daniel Lidar, K. B. Whaley, "Universal Fault-Tolerant Computation on Decoherence-Free Subspaces" quant-ph/9909058. (1999)

Julia Kempe, Dave Bacon, Daniel Lidar, K. B. Whaley, "Theory of Decoherence-Free Fault-Tolerant Universal Quantum Computation." quant-ph/0004064. (2000)

David DiVincenzo, Guido Burkard, Julia Kempe, Dave Bacon, work in progress (2000)

Heisenberg Coupling

Bob



Al, I'm still waiting for your answer? Can I quantum compute?

Look at those spin-coupled quantum dots you have. What is the coupling between the dots?

Al



$H_{heis} = J(t)\, s^{(i)} \cdot s^{(j)}$ where $s^{(i)} = \left(s_x^{(i)}, s_y^{(i)}, s_z^{(i)}\right)$ are the single spin operators on qubit i

$$s^{(i)} \cdot s^{(j)} = s_x^{(i)} s_x^{(j)} + s_y^{(i)} s_y^{(j)} + s_y^{(i)} s_y^{(j)} = \tfrac{1}{4}\left(E_{ij} - I\right)$$

since I in Hamiltonian does not matter (global phase):

the Heisenberg Coupling is exactly the Exchange Coupling of quantum dot / Kane proposal

Not necessary to supplement the Heisenberg coupling with single qubit operations!

- Heisenberg interaction in quantum dot / Kane proposal are easily tuned using local magnetic fields.
- Heisenberg operation is fast (GHz range) and has an on-off ratio of several orders of magnitude.
- Encoding into low J DFSs is easily performed using a simple cooling routine: let there be an exchange interaction between neighboring dots and cool the system.

Many more details of the application of this simple universality result have been worked out and are being detailed in work with David DiVincenzo, Guido Burkard, and Julia Kempe.

A Duality

For DFSs, since the decoherence avoidence is independent of the coupling strength, in order to perform fault-tolerant computation for strong coupling,

computation should NEVER force the states outside of the DFS.

This is true for the exchange interaction. *Naturally fault tolerant!*

Contrast standard independent error quantum error correction (QEC). In this case any *single or two body gate MUST take the state outside of the code* during the operation time of the gate:

U(t)=exp[iHt] for small t = I+iHt but if H is single or two body interaction

then I+iHt will be an ERROR on the QEC and so will not preserve the subspace.

The reason for this is a nice DUALITY:

• collective error model has errors which are global so local operations can preserve the subspace

• independent error model has errors which are local so global operations are needed to preserve the subspace

What Do I Need To Quantum Compute?



Hey I found this new quantum system with these crazy interactions, Al. Can I quantum compute?

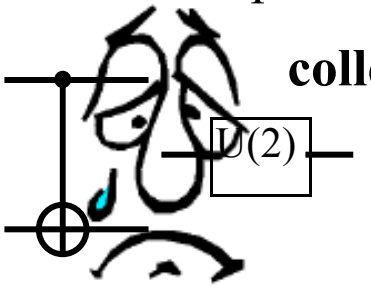
Many interactions, with a bit of encoding, can be used to do quantum computation.

This question is a simple application of the representation theory of Lie Algebras (take it to your local particle theorist for a quick answer!)

A few more qubits and a few more operations may be required.

The hope, however, is that the requirements for engineering a quantum computer using encoded universal interactions will be simplified.

Fault-tolerant computation on collective DFSs is possible!



Controlled-Not and single qubit gates are not the only way to go!



Acknowledgements

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Dr. Daniel Lidar (UC Berkeley, Chemistry, soon University of Toronto, Chemistry)

Professor K. B. Whaley (UC Berkeley, Chemistry)

Dr. David DiVincenzo (IBM)

Guido Burkard (University of Basel)

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Dave Bacon, Julia Kempe, Daniel Lidar, K. B. Whaley, "Universal Fault-Tolerant Computation on Decoherence-Free Subspaces" quant-ph/9909058. (1999)

Julia Kempe, Dave Bacon, Daniel Lidar, K. B. Whaley, "Theory of Decoherence-Free Fault-Tolerant Universal Quantum Computation" quant-ph/0004064. (2000)

Daniel Lidar, Dave Bacon, K. B. Whaley "Concatenating Decoherence Free Subspaces with Quantum Error Correcting Codes," Phys. Rev. Lett. **82**, 4556. quant-ph/9809081. (1999)